

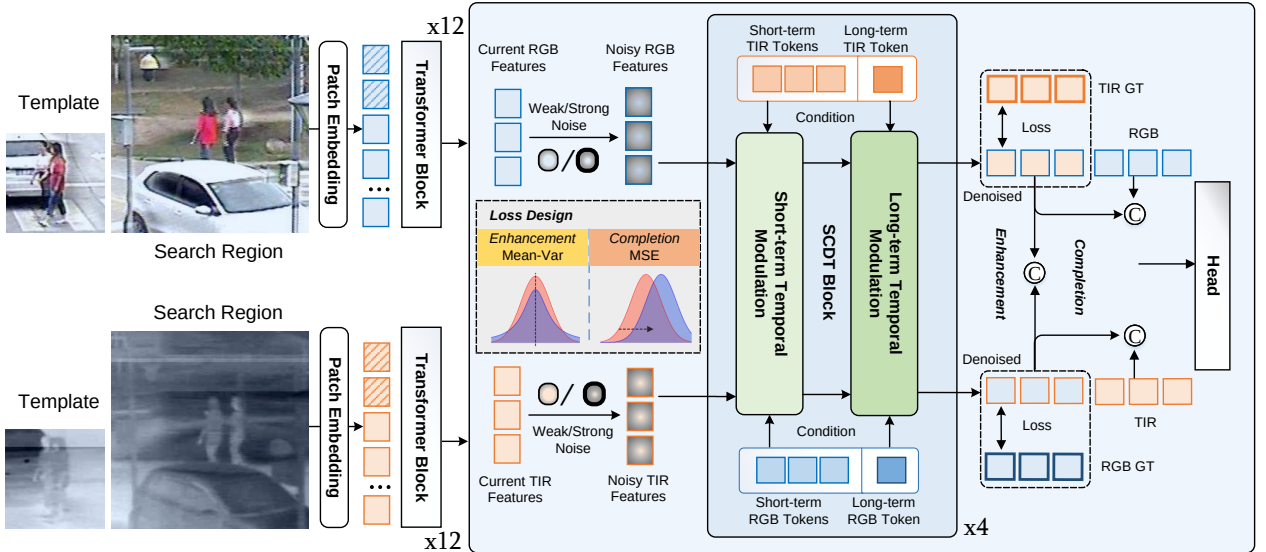


Figure 2. Overall architecture of the proposed SCDT. SCDT performs a noising-denosing process on available search frames, guided by historical frame features with short-term and long-term temporal dependencies. This enables simultaneous feature enhancement for complete modalities and reconstruction for missing modalities within a unified framework.

Modality Feature Enhancement. When both modalities are available, SCDT follows the same conditional generation pathway under a weak-noise regime. Instead of enforcing pixel-wise fidelity, we encourage the model to generate an *enhanced* feature $\hat{f}_m$ that remains close to the true distribution while offering improved discriminability. To this end, we align the first- and second-order statistics of the generated and real features:

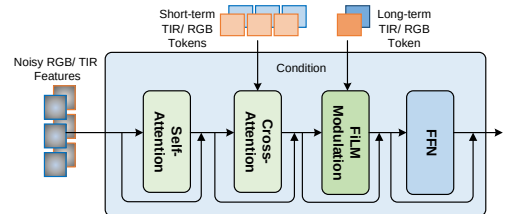


Figure 3. Detailed architecture of the Short-term and Long-term Temporal Modulation modules.